

Spreading 38 Points in the Plane

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Abstract

We place 38 points in the plane so that the squared ratio of the largest to the smallest distance between them is at most 35.26616+. Erich Friedman’s table of best known configurations lists n up to 31. Deleting a point from our 39-point configuration gives 36.28378; this one is smaller. The ratio is computed exactly from the published coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 38 points in the plane with*

$$\left(\frac{\max d}{\min d}\right)^2 = 35.26616\dots$$

The exact value, a rational number, is given in Appendix A.

2 Context

Erich Friedman’s page *Minimizing the Ratio of Maximum to Minimum Distance* lists the best known configurations for $n \leq 31$. The value for $n = 38$ is not on the page. Deleting a point from our 39-point configuration gives 36.28378; this one is smaller.

3 Verification

The coordinates are decimals with 12 digits after the point, so they are exact rationals; all squared distances, and their largest-to-smallest ratio, are computed with integers. Run

```
python friedman-maxmin2d/verify.py friedman-maxmin2d/coords/n38.txt
```

in <https://github.com/Ge-limin/math>.

4 How it was found

Random starts of a constrained local optimizer (SLSQP), then basin hopping seeded from the neighbouring sizes (`friedman-maxmin2d/`).

A Exact value

```
numerator:
7053232277695242993630425
denominator:
199999999999643402739853
```

*Found and checked by Claude, a model released by Anthropic, running on the author’s laptop. The author sent the prompts.

B Coordinates

38 points, x and y per line. File: friedman-maxmin2d/coords/n38.txt.

```
-2.595786338634 5.456246616298 -1.265302569796 6.879177418168
-1.571963427700 1.873507425457 0.243235203037 4.536402181023
-3.979237687296 6.878819030561 -1.626176870886 7.811791882356
-3.676317430283 4.934853256995 -4.205142023253 3.122536573355
-0.974743060858 3.057648369256 -4.479087341999 6.012706842277
-1.260653230217 5.772539241497 -4.671643848907 5.031420955470
-0.605021896805 2.128505613351 0.803563862818 6.312995276318
-1.701605134236 4.807905488606 0.197211551297 7.108191399493
-3.473957481593 2.440356856537 -2.015690508579 3.829055362004
1.164438163908 5.380380812130 -2.859141676464 3.291849509986
1.231340171505 4.382621261246 -0.430019116082 6.329357855082
-2.983911268672 6.782251332086 -1.041645068455 4.055407920140
-3.623848733876 3.936230687704 -0.638071902417 7.658010962579
-2.780396444137 4.473434778282 -2.095936683931 6.322358804582
-2.621406655812 7.714233309194 0.225612217331 2.685324226936
-0.144108946723 3.614466982841 0.869638988579 3.450327179816
-3.483760923375 5.916139143802 0.176333195439 5.534161731907
-0.654300918696 4.977343118322 -2.558260241191 2.038488017336
-1.948788500981 2.831295811121 -4.619175152500 4.032798386179
```